JORDAN ARYA LEKSANA

Software Engineer · Full-Stack & AI

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SUMMARY

Software Engineer with an Engineering Physics background, focused on full-stack development and production deployment. Experienced in building end-to-end web, AI-powered and IoT applications using TypeScript, Next.js, Python (FastAPI, Flask), Node.js, REST APIs, PostgreSQL and Redis, and in shipping them with Docker, GitHub Actions CI/CD and Linux-based infrastructure. Skilled in integrating LLMs, RAG and machine learning models into real products, from a solo-built AI video editor for the Meta Global AI Developer Hackathon 2026 to a live RAG portfolio agent.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL; Go (basic)

Frontend & Mobile: React, Next.js, Tailwind CSS, React Native

Backend & APIs: Node.js, Express, NestJS, FastAPI, Flask, REST APIs, NextAuth, Prisma, SQLAlchemy

Databases: PostgreSQL, MySQL, MongoDB, Redis

DevOps & Cloud: Docker, Docker Compose, GitHub Actions (CI/CD), GHCR, Nginx, Linux (Ubuntu) VPS, Vercel, Cloudflare

AI / ML: LLM integration (OpenAI, Anthropic, Ollama), Vercel AI SDK, LangChain, RAG, PyTorch, TensorFlow, scikit-learn, OpenCV

Tools: Git, Postman, Agile/Scrum

EXPERIENCE

PT Pertamina EP Cepu — Cepu, Central Java

AI Engineer Intern — Jan 2025 – May 2025

- Built an LSTM-Autoencoder anomaly-detection pipeline in Python/TensorFlow on multivariate sensor data from three industrial solvent pumps, flagging early signs of mechanical-seal failure; contributed to zero lost production hours on the monitored pump.
- Processed 5,000+ anomaly events, validated them with field operators, and trained a supervised classifier that categorized them at 85% validation accuracy.
- Turned pump specifications and operator knowledge into a rule-based early-warning and prescriptive-maintenance layer, with estimated potential cost savings of up to USD 476K.
- Contributed to the methodology, experiments and manuscript later presented as SPE-224966-MS (LEADS).

Embedded System and Cyber-Physical Laboratory (ECS), ITS — Surabaya, Indonesia

Deep Learning Researcher — Dec 2025 – Jul 2026

- Built a ResNet-based PyTorch pipeline that detects geomagnetic earthquake precursors across 25,000+ events from 24 stations in Indonesia, reaching 88.25% accuracy and 88.4% F1 and automating a manual analysis method.

Software Development Lead — Jun 2025 – Nov 2025

- Designed the end-to-end architecture of the TETA smart-farming system: React Native mobile app, backend server, ESP32–LoRa–MQTT communication and Raspberry Pi edge devices.
- Shipped it to a 1-hectare rice field in Tuban, East Java (5 IoT nodes): real-time pH, NPK, EC, temperature and humidity monitoring plus 24/7 YOLOv5 computer-vision bird-pest deterrence, reducing fertilizer waste and manual field guarding.
- Wrote the user guide (registered copyright No. 001036674) so non-technical farmers can run the system on their own.

IT Project Manager — Jan 2025 – Jul 2025

- Led delivery of ECS Website 2.0 and the EPTA event dashboard with a cross-functional team of developers and designers, owning scope, sprint planning and timelines in an Agile process.

Frontend Tech Lead — Jun 2024 – Nov 2024

- Led frontend development of the ECS website and the EPTA event portal in Next.js and Tailwind CSS, serving 100+ event participants.

- Built reusable UI components and integrated REST APIs for authentication, registration, module and certificate uploads, assignments and task submissions.

PROJECTS

IntentCut — AI Video Editor | Next.js, React, TypeScript, FastAPI, PostgreSQL, SQLAlchemy, FFmpeg GitHub · Demo

- Built solo in 10 days for the Meta Global AI Developer Hackathon 2026: generates a first cut and highlight reels from long footage using Meta's Muse Spark model.
- Designed a background job worker with database leases, heartbeats and crash recovery; the backend validates every AI-proposed edit against real source timestamps before rendering.
- Implemented captions and H.264/AAC MP4 export with FFmpeg; in testing, turned a 4-hour, 6.4 GB recording into a ~56-second highlight reel.

Jordan Labs — AI Portfolio Agent | Next.js, TypeScript, Vercel AI SDK, Redis, Docker, Nginx jordanlabs.my.id

- Built a portfolio site whose assistant answers questions with hybrid RAG over verified data and uses schema-validated tool calling to navigate pages and assess role fit from a job description.
- Added production safeguards: Redis rate limiting, daily request quotas and token-budget tracking to control LLM cost.
- Set up CI/CD with GitHub Actions and GHCR that tests, builds and deploys Docker images to a Linux VPS behind Nginx and HTTPS.

SEA Catering — Meal Subscription Platform | Next.js, TypeScript, Prisma, NextAuth, Vercel Live

- Built a full-stack subscription app: meal plans, subscription create/pause/cancel flows and delivery schedules.
- Implemented email and Google sign-in with NextAuth, role-based user and admin dashboards with business metrics, and validated API routes over a Prisma data model.

DermSight — AI Skin Screening Web App | Python, Flask, TensorFlow, OpenCV, Ollama, Docker Live

- Shipped a mobile-first web app that classifies 22 skin conditions with EfficientNet-B2 and explains results with a local LLM grounded by RAG.
- Hardened it with CSRF protection, image validation, upload limits and security headers; photos are processed in memory and never stored. Deployed with Docker and GitHub Actions to a Linux VPS.

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS) Surabaya, Indonesia

Bachelor of Engineering in Engineering Physics · GPA 3.28/4.00 2022 – Oct 2026

- Thesis: EMQNET, multi-task deep learning for earthquake precursor detection from geomagnetic data.
- Relevant courses: Computer Programming, Logic and Digital Systems, Engineering Mathematics, Artificial Neural Networks.

PUBLICATIONS, AWARDS & ACTIVITIES

- **Meta Global AI Developer Hackathon 2026** — solo participant (Indie track) with IntentCut.
- **SPE-224966-MS** — LEADS: Leading Equipment Anomaly Detection System. Research and technical-writing contributor.
- **A CNN-Based Approach for Earthquake Precursor Detection Using Geomagnetic Data** — first author, manuscript in preparation (2026).
- **IoT-Based Soil Quality and Pest Monitoring for Sustainable Farming (Gapoktan Manunggal Rejeki)** — co-author, BIMA/ITS community-service publication (2025).

LANGUAGES

English (Fluent) · Indonesian (Native)